beamer named overlay specifications with beanoves

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Abstract

This package allows the management of multiple named overlay specifications in beamer documents. Named overlay specifications are very handy both during edition and to manage complex and variable beamer overlay specifications. In particular, they allow to replace raw numbers in beamer <...> overlay specifications by logical identifiers. Demonstration files are available for download as part of the development repository. As a side effect, this is another solution to this latex.org forum query.

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1.1 Package declarations

```
\NeedsTeXFormat{LaTeX2e}[2025/01/01]
\ProvidesExplPackage
  {beanoves}
  {2025/06/03}
  {1.0}
  {Named overlay specifications for beamer}
```

1.2 Overview

Reserved namespace: identifiers containing the case insensitive string `beanoves` or containing the case insensitive string `bnvs` delimited by two non characters.

There are mainly two steps: parsing and resolution. Parsing occurs while executing the `\Beanoves` command to build a data model whereas the resolution translates queries into beamer specifications based on this data model.

The development is made easier due to a facility layer over expl.

```
\regex_const:Nn \c__bnvs_A_VAZLLZZL_Z_regex {
  \A \s* (?:
```

- 2 → V

```
    ( [^:]+? )
```

- 3, 4, 5 → A : Z? or A :: L?

```
    | (?: ( [^:]+? ) \s* : (?: \s* ( [^:]*? ) | : \s* ( [^:]*? ) ) )
```

- 6, 7 → ::(L:Z)?

```
    | (?: :: \s* (?: ( [^:]+? ) \s* : \s* ( [^:]+? ) )? )
```

- 8, 9 → :(Z::L)?

```
    | (?: : \s* (?: ( [^:]+? ) \s* :: \s* ( [^:]*? ) )? )
  )
  \s* \Z
}
```

1.3 Main scanner

We use the same scanner to parse definitions and queries. We do not use any key–value facility built into LaTeX.

1.3.1 Base grammar

Hereafter is a pragmatic grammar used for parsing, not all the expressions covered do have a real meaning. The characters all have category other. We start with basic entities.

```
⟨spc⟩          ::= Horizontal white space characters
⟨lower⟩        ::= a lowercase letter
⟨upper⟩        ::= an uppercase letter
⟨digit⟩        ::= a decimal digit
⟨sign⟩         ::= '-' | '+'
⟨comma⟩        ::= ⟨spc⟩ ',' ⟨spc⟩
⟨alpha⟩        ::= ⟨lower⟩ | ⟨upper⟩ | '_'
⟨alnum⟩        ::= ⟨alpha⟩ | ⟨digit⟩
⟨index⟩        ::= ⟨sign⟩* ⟨digit⟩+
⟨number⟩       ::= ⟨sign⟩? ⟨significand⟩ ⟨exponent⟩?
⟨significand⟩  ::= ( '.'? ⟨digit⟩+ ) | ( ⟨digit⟩+ '.' ⟨digit⟩* )
⟨exponent⟩     ::= ( 'e' | 'E' ) ⟨sign⟩? ⟨digit⟩+
⟨other⟩        ::= ⟨number⟩ | '+' | '-' | '*' | '/'
```

\c__bnvs_N_regex Signed integer.

(*End of definition for* \c__bnvs_N_regex.)

```
\regex_const:Nn \c__bnvs_N_regex {
  [-+]* \d+
}
```

\c__bnvs_dot_N_regex Signed integer with a dot . just before.

(*End of definition for* \c__bnvs_dot_N_regex.)

```
\regex_const:Nn \c__bnvs_dot_N_regex {
  \. \ur{c__bnvs_N_regex}
}
```

\c__bnvs_A_dot_N_Z_regex Full signed integer with a dot . jus before.

(*End of definition for* \c__bnvs_A_dot_N_Z_regex.)

```
\regex_const:Nn \c__bnvs_A_dot_N_Z_regex {
  \A \ur { c__bnvs_dot_N_regex } \Z
}
```

\c__bnvs_A_i_Z_regex Signed integer with no capture groups.

(*End of definition for* \c__bnvs_A_i_Z_regex.)

```
\regex_const:Nn \c__bnvs_A_i_Z_regex { \A \ur{c__bnvs_N_regex} \Z }
```

\c__bnvs_dot_R_regex A possibly empty list of .⟨*short name*⟩ items representing a path. Integers are allowed as well.

```
\regex_const:Nn \c__bnvs_R_regex {
  \ur{c__bnvs_N_regex} | [[:alnum:]_]+
}
```

(*End of definition for* \c__bnvs_dot_R_regex.)

\c__bnvs_dot_R_regex A possibly empty list of .⟨*short name*⟩ items representing a path. Integers are allowed as well.

```
\regex_const:Nn \c__bnvs_dot_R_regex {
  \. \ur{c__bnvs_R_regex}
}
```

(*End of definition for* \c__bnvs_dot_R_regex.)

\c__bnvs_S_regex This regular expression is used for short names. The short name of an overlay set consists of a non void list of alphanumerical characters and underscore, but with no leading digit.

```
\regex_const:Nn \c__bnvs_S_regex {
  [[:alpha:]_][[:alnum:]_]*
}
```

(*End of definition for* \c__bnvs_S_regex.)

\c__bnvs_P_regex This regular expression is used for short alphanumerical path components with at least one letter.

```
\regex_const:Nn \c__bnvs_P_regex {
  \d*[[:alpha:]_][[:alnum:]_]*
}
```

(*End of definition for* `\c__bnvs_P_regex`.)

`\c__bnvs_dot_P_regex` This regular expression is used for short alphanumerical path components with at least one letter.

```
\regex_const:Nn \c__bnvs_dot_P_regex {
  \. \ur{c__bnvs_P_regex}
}
```

(*End of definition for* `\c__bnvs_dot_P_regex`.)

1.3.2 Raw atoms

```
⟨Raw⟩    ::= '"' '(' ⟨RawLst⟩ ')'
⟨RawLst⟩ ::= ( ⟨spc⟩ | ⟨Rlv⟩ | ⟨RawRnd⟩ | ⟨RawSqr⟩ | ⟨RawCrl⟩ | ⟨RawOtr⟩ )*
⟨RawRnd⟩ ::= '(' ⟨RawLst⟩ ')'
⟨RawSqr⟩ ::= '[' ⟨RawLst⟩ ']'
⟨RawCrl⟩ ::= '{' ⟨RawLst⟩ '}'
⟨RawOtr⟩ ::= ⟨comma⟩ | ⟨alnum⟩ | ⟨other⟩
```

1.3.3 Storage

There are different policies to store the intermediate results of a scanning operation. One possibility is to feed a global variable as scanning occurs. As you cannot use unbalanced braces in function arguments, we can use other characters than usual "\", "{" and "}" before we rescan the final result with an appropriate catcode regime. Another possibility is to heavily use TEX groups to store intermediate results in local variables and only consider properly balanced material. At the end of the group, the scanned material must be forwarded to the group owner. The latter is used here.

There is also a special situation for raw material that is stored separately and exported at once before other material is exported or when a scan group ends.

`\BNVS_xprt_clear:` `\BNVS_xprt_clear:`
`\BNVS_xprt:n` `\BNVS_xprt:n {⟨material⟩}`
`\BNVS_scan_xprt_black:n` `\BNVS_scan_xprt_black:n {⟨material⟩}`

`\BNVS_xprt_clear:` resets the scan storage, called when starting a new TEX group dedicated to scanning. `\BNVS_xprt:n` is a utility function to store its argument in the `xprted` variable. Automatically set by `\BNVS_begin_scan:...` functions. Used by `\BNVS_end_scan:`. The default implementation just raises. `\BNVS_xprt_grp:n` feeds the scan storage with ⟨*material*⟩, enclosed betwenn one pair of curly braces. `\BNVS_scan_xprt_black:n` is called just before exporting. The argument is enclosed inside `\exp_not:n{⟨...⟩}`. The default implementation just expands to its argument as is.

```
\cs_new:Npn \BNVS_xprt_clear: {
  \__bnvs_tl_clear:c { xprted }
}
```

```
\cs_new:Npn \BNVS_xprt:n #1 {
  \BNVS_xprt_stored:
  \__bnvs_tl_put_right:cn { xprted } { #1 }
}
```

\BNVS_xprt_grp:n `\BNVS_xprt_grp:n {⟨scanned⟩}`
\BNVS_xprt_use:n `\BNVS_xprt_use:n {⟨code:n⟩}`

`\BNVS_xprt_grp:n` exports its argument between curly braces. `\BNVS_xprt_use:n` puts back to the input stream the exported material so far, enclosed in one pair of braces after the instructions ⟨*code:n*⟩.

```
\cs_new:Npn \BNVS_xprt_use:n #1 {
  \BNVS_xprt_stored:
  \BNVS_tl_use:nv { #1 } { xprted }
}

\cs_new:Npn \BNVS_xprt_grp:n #1 {
  \BNVS_xprt:n { { #1 } }
}
```

\BNVS_if_xprted_empty:TF `\BNVS_if_xprted_empty:TF {⟨yes:n⟩} {⟨no:n⟩}`

Execute ⟨*yes:n*⟩ instructions if something has been exported, ⟨*no:n*⟩ instructions otherwise.

```
\cs_new:Npn \BNVS_if_xprted_empty:TF {
  \BNVS_xprt_use:n { \tl_if_empty:nTF }
}
```

\BNVS_xprt_wrap_prefix:n `\BNVS_xprt_wrap_prefix:n {⟨xprt prefix:w⟩}`

Set the function `\BNVS_xprt_prefix:w` to expand to `\exp_not:n { ⟨xprt prefix:w⟩}`. This is used to define `\BNVS_xprt_wrapped:n`, when the arguments must be scanned prior to knowing which function ⟨*wrapped:w*⟩ fits.

```
\cs_new:Npn \BNVS_xprt_prefix:w {}

\cs_new:Npn \BNVS_xprt_wrap_prefix:n #1 {
  \cs_set:Npn \BNVS_xprt_prefix:w { \exp_not:n { #1 } }
}
```

\BNVS_xprt_wrap:n `\BNVS_xprt_wrap:n {⟨xprt:n⟩}`
\BNVS_xprt_wrap_grp:n `\BNVS_xprt_wrap_grp:n {⟨xprt:n⟩}`
\BNVS_xprt_wrap_raw:n `\BNVS_xprt_wrap_raw:n {⟨xprt:n⟩}`

Set the function `\BNVS_xprt_wrapped:n`. The first variant exports only black prepared material. The second variant exports material between curly brackets. The third variant exports material a is.

```
\cs_new:Npn \BNVS_xprt_wrap:n #1 {
  \cs_set:Npn \BNVS_xprt_wrapped:n ##1 {
    \tl_if_blank:nF { ##1 } {
      \BNVS_xprt_prefix:w
```

and `\exp_not:n{⟨xprt:n⟩}`.

```
      \exp_not:n { #1 }
    }
  }
}
\BNVS_xprt_wrap:n { #1 }

\cs_new:Npn \BNVS_xprt_wrap_raw:n #1 {
  \cs_set:Npn \BNVS_xprt_wrapped:n ##1 {
    \BNVS_xprt_prefix:w
    \tl_if_blank:nF { ##1 } {
```

and `\exp_not:n{⟨xprt:n⟩}`.

```
      \exp_not:n { #1 }
    }
  }
}

\cs_new:Npn \BNVS_xprt_wrap_grp:n #1 {
s
  \cs_set:Npn \BNVS_xprt_wrapped:n ##1 {
    \BNVS_xprt_prefix:w
```

and `\exp_not:n{⟨xprt:n⟩}`.

```
    \exp_not:n { { #1 } }
  }
}
```

`\BNVS_xprt_wrap_use:n` `\BNVS_xprt_wrap_use:n {⟨code:n⟩}`

It applies ⟨*code:n*⟩ to the currently exported material, after it is wrapped..

```
\cs_new:Npn \BNVS_xprt_wrap_use:n #1 {
  \BNVS_xprt_stored:
  \exp_args:Nnx \use:n { #1 } {
    \BNVS_tl_use:nv { \BNVS_xprt_wrapped:n } { xprted }
  }
}
```

`\BNVS_xprt_store:n` `\BNVS_xprt_store:n {⟨something⟩}`
`\BNVS_xprt_stored:` `\BNVS_xprt_stored:`

`\BNVS_xprt_store:n` synchronously cumulates its argument in the `stored` variable. The function `\BNVS_xprt_stored:` is executed both at the beginning of the function `\BNVS_xprt:n` and when a scan group ends. It exports the contents of the stored ⟨*material*⟩, when non empty, to the `xprted` variable as `\:n{⟨material⟩}` instruction and cleans the `stored` variable.

```
\cs_new:Npn \BNVS_xprt_stored: {
  \__bnvs_tl_if_empty:cF { stored } {
    \BNVS_tl_use:nv {
      \__bnvs_tl_clear:c { stored }
      \BNVS_xprt:n { \:n }
      \BNVS_xprt_grp:n
    } { stored }
  }
}
```

```
\cs_new:Npn \BNVS_xprt_store:n #1 {
  \tl_if_empty:nF { #1 } {
    \__bnvs_tl_put_right:cn { stored } { #1 }
  }
}
```

\BNVS_next_set:n `\BNVS_next_set:n {⟨next:⟩}`
\BNVS_next_use:n `\BNVS_next_use:n {⟨code:⟩}`

`\BNVS_next_set:n` stores ⟨*next:*⟩ to be executed later by `\BNVS_next_use:n`, just after ⟨*code:*⟩. Typically, `\BNVS_next_set:n` is used in a scan group closed by ⟨*code:*⟩. `\BNVS_end_scan_next:` ends the scan and executes ⟨*next:*⟩ code.

```
\BNVS_tl_new:c { next }
\cs_new:Npn \BNVS_next_set:n #1 {
  \__bnvs_tl_set:cn { next } {
    \exp_not:n {
      #1
    }
  }
}

\cs_new:Npn \BNVS_next_use:n #1 {
  \BNVS_tl_use_unbraced:nv { #1 } { next }
}
```

1.3.4 Close branch

The scanning process only applies to characters of category codes space and other. It definitely stops at the first token that is not of that category. When encontering some particular character combinations, various branch functions are called.

\c__bnvs_close_rnd_regex Round closing delimiter.

```
\regex_const:Nn \c__bnvs_close_rnd_regex { \s* %(
  \) \s* }
```

(*End of definition for* `\c__bnvs_close_rnd_regex`.)

\c__bnvs_close_sqr_regex Square closing delimiter.

```
\regex_const:Nn \c__bnvs_close_sqr_regex { \s* %[
  \] \s* }
```

(*End of definition for* `\c__bnvs_close_sqr_regex`.)

\c__bnvs_close_crl_regex Curly closing delimiter.

```
\regex_const:Nn \c__bnvs_close_crl_regex { \s* %{
  \} \s* }
```

(*End of definition for* `\c__bnvs_close_crl_regex`.)

\c__bnvs_close_rnd_regex Round closing delimiter.

```
\regex_const:Nn \c__bnvs_close_regex { \s* (%([{
  \}|\]|\)) \s* }
```

(*End of definition for* `\c__bnvs_close_rnd_regex`.)

`\BNVS_scan_close:Fw`	`\BNVS_scan_close:Fw {`⟨*else:*⟩`}` ⟨*input stream*⟩
`\BNVS_scanned_close:Nw`	`\BNVS_scanned_close:Nw` ⟨*clositer*⟩ ⟨*input stream*⟩
`\BNVS_scanned_close_set:`	`\BNVS_scanned_close:w` ⟨*input stream*⟩
`\BNVS_scanned_close:w`	`\BNVS_scanned_close_set:n` ⟨*close:w*⟩

When a closing delimiter is scanned, call `\BNVS_scanned_close:w` with that delimiter as argument. Otherwise execute ⟨*else:*⟩ code.

`\BNVS_scanned_close:w` is a branch function called when the closing delimiter ⟨*clositer*⟩ has just been scanned. After it manages eventual errors in balancing delimiters, it calls the branch function `\BNVS_scanned_close:w` used many times. When redefined (for example by `\__bnvs_scan_delimited:NNnFw`), the `\BNVS_scanned_close:w` function does not forward to the eponym function, instead it must know where to go next.

```
\cs_new:Npn \BNVS_scan_close:Fw #1 {
  \peek_regex_replace_once:NnTF \c__bnvs_close_regex { \1 } {
    \BNVS_scanned_close:Nw
  } {
    #1
  }
}

\cs_new:Npn \BNVS_scanned_close:Nw #1 {
  \BNVS_error:n { Unexpected~delimiter~`#1' }
  \BNVS_scanned_close:w
}

\tl_map_inline:nn {{close}{comma}{one_colon}{colons}{alter}} {
  \cs_new:cpn { BNVS_scanned_#1_set:n } ##1 {
    \cs_set:cpn { BNVS_scanned_#1:w } {
      ##1
    }
  }
```

Do nothing operation at the top level.

```
  \cs_new:cpn { BNVS_scanned_#1:w } {
  }
}
```

1.3.5 Comma branch

`\c__bnvs_comma_regex` Comma as a separator.

```
\regex_const:Nn \c__bnvs_comma_regex { \s* , \s* }
```

(*End of definition for* `\c__bnvs_comma_regex`.)

`\BNVS_scan_comma:Fw`	`\BNVS_scan_comma:Fw {`⟨*else:w*⟩`}` ⟨*input stream*⟩
`\BNVS_scanned_comma:w`	`\BNVS_scan_comma:w` ⟨*input stream*⟩
`\BNVS_scanned_comma_set:n`	`\BNVS_scan_comma_set:n {`⟨*comma:w*⟩`}`

Once a comma has just been scanned from the head of the ⟨*input stream*⟩, calls `\BNVS_scanned_comma:w` branch function, otherwise execute ⟨*else:w*⟩ instructions. `\BNVS_scan_comma_set:n` sets the meaning of `\BNVS_scan_comma:w` to ⟨*comma:w*⟩.

```
\cs_new:Npn \BNVS_scan_comma:Fw #1 {
```
\peek_regex_remove:NnTF is buggy in L3 programming layer <2025-01-18>.
```
  \peek_regex_replace_once:NnTF \c__bnvs_comma_regex {} {
    \BNVS_scanned_comma:w
  } {
    #1
  }
```
There must be absolutely no code here. No hooking allowed either.
```
}
```

1.3.6 Colon branch

\c__bnvs_colons_regex For ranges defined by a colon syntax. One catching group for more than one colon.

```
\regex_const:Nn \c__bnvs_colons_regex { \s* :(:*) \s* }
```

(*End of definition for* \c__bnvs_colons_regex.)

\BNVS_scan_colon:Fw	\BNVS_scan_colon:Fw {⟨*else:*⟩} ⟨*input stream*⟩
\BNVS_scanned_one_colon:w	\BNVS_scan_colon:Fw {⟨*else:w*⟩} ⟨*input stream*⟩
\BNVS_scanned_one_colon_set:n	\BNVS_scan_one_colon:w ⟨*input stream*⟩
\BNVS_scanned_colons:w	\BNVS_scan_one_colon_set:n {⟨*one_colon:w*⟩} \BNVS_scan_colons:w ⟨*input stream*⟩
\BNVS_scanned_colons_set:n	\BNVS_scan_colons_set:n {⟨*colons:w*⟩}

On scanning a standalone colon from the ⟨*input stream*⟩, calls the branch function \BNVS_scanned_one_colon:w. On scanning multiple colons from the ⟨*input stream*⟩, calls the \BNVS_scanned_colons:w branch function. In other cases, execute the ⟨*else:w*⟩ instructions.

One shot utility function only used by \BNVS_scan_colon:Fw

```
\cs_new:Npn \BNVS_scanned_colons_regex:n #1 {
  \tl_if_empty:nTF { #1 } {
    \BNVS_scanned_one_colon:w
  } {
    \BNVS_scanned_colons:w
  }
}

\cs_new:Npn \BNVS_scan_colon:Fw #1 {
  \peek_regex_replace_once:NnTF \c__bnvs_colons_regex { { \1 } } {
    \BNVS_scanned_colons_regex:n
  } {
    #1
  }
}
```

1.3.7 Stop branch

\BNVS_scanned_stop:
\BNVS_scanned_stop_set:n `\BNVS_scanned_stop_set:n {⟨stop:⟩}`

The `\BNVS_scanned_stop:` branch function is called when there is nothing more to scan. `\BNVS_scanned_stop_set:n` sets the meaning of `\BNVS_scanned_stop:` to ⟨*stop:*⟩.

```
\cs_new:Npn \BNVS_scanned_stop_set:n #1 {
  \cs_set:Npn \BNVS_scanned_stop: {
    #1
  }
}
```

Do nothing operation at the top level.

```
\cs_new:Npn \BNVS_scanned_stop:w {
}
```

\BNVS_scan_stop:F `\BNVS_scan_stop:F {⟨else:n⟩} ⟨input stream⟩`

On scanning what is not space nor a character of category code other, calls the `\BNVS_scanned_stop:` branch function, otherwise execute ⟨*else:n*⟩ instructions.

```
\cs_new:Npn \BNVS_scan_stop:F #1 {
  \peek_catcode:NTF \c_catcode_other_token { #1 } {
    \peek_catcode_remove:NTF \c_space_token {
      \BNVS_xprt_store:n { ~ }
      #1
    } {
      \BNVS_scanned_stop:
    }
  }
```

There must be absolutely no code here. No hooking allowed either.

```
}
```

1.3.8 Alter branch

The alter branch may vary.

\BNVS_scanned_alter:w `\BNVS_scanned_alter:w ⟨input stream⟩`
\BNVS_scanned_alter_set:n `\BNVS_scanned_alter_set:n {⟨alter:w⟩}`

The `\BNVS_scanned_alter:w` branch function is called by `\BNVS_scan_alter:Fw` once ⟨*alter*⟩ has been scanned from the head of the ⟨*input stream*⟩ (greedy match). `\BNVS_scanned_alter_set:n` sets the meaning of `\BNVS_scanned_alter:w` to ⟨*alter:w*⟩.

\BNVS_scan_alter:Fw `\BNVS_scan_alter:Fw {⟨else:w⟩} ⟨input stream⟩`
\BNVS_scan_alter:TFw `\BNVS_scan_alter:TFw {⟨yes:w⟩} {⟨else:w⟩} ⟨input stream⟩`
\BNVS_scan_alter_wrap:n `\BNVS_scan_alter_wrap:n {⟨alter:TFw⟩}`

Once ⟨*alter*⟩ has been scanned from the head of the ⟨*input stream*⟩ (greedy), calls `\BNVS_scanned_alter:w` branch function, otherwise execute ⟨*else:w*⟩ instructions. Scanning ⟨*alter*⟩ is driven by ⟨*alter:TFw*⟩.

```
\cs_new:Npn \BNVS_scan_alter:Fw #1 {
```

\peek_regex_remove:NnTF is buggy in L3 programming layer <2025-01-18>.

```
  \BNVS_scan_alter:TFw {
    \BNVS_scanned_alter:w
  } {
    #1
  }
```

There must be absolutely no code here. No hooking allowed either.

```
}

\cs_new:Npn \BNVS_scan_alter:TFw #1 #2 {
```

At the top level, the ⟨*else:w*⟩ instructions are always executed.

```
  #2
}

\cs_new:Npn \BNVS_scan_alter_wrap:n #1 {
  \cs_set:Npn \BNVS_scan_alter:TFw ##1 ##2 {
    #1
  }
}
\BNVS_scan_alter_wrap:n { #2 }
```

1.3.9 Control functions

\BNVS_scan_delimiter:Fw `\BNVS_scan_delimiter:Fw {⟨else:⟩} ⟨input stream⟩`

Medley of the above,

```
\cs_new:Npn \BNVS_scan:w {
  \BNVS_scan_close:Fw {
    \BNVS_scan_comma:Fw {
      \BNVS_scan_colon:Fw {
        \BNVS_scan_stop:F {
          \BNVS_scan_alter:Fw {
            \BNVS_xprt_store:N
          }
        }
      }
    }
  }
}

\cs_new:Npn \BNVS_scan_delimiter:Fw #1 {
  \BNVS_scan_close:Fw {
    \BNVS_scan_comma:Fw {
      \BNVS_scan_colon:Fw {
        \BNVS_scan_stop:F {
          #1
        }
      }
    }
```

```
  }
}
```

1.3.10 Control functions

\BNVS_scan:nnnnnnnnw
\BNVS_scan:nnw

```
\BNVS_scan:nnnnnnnnw {⟨preamble:⟩}
  {⟨closed:⟩} {⟨comma:⟩} {⟨one_colon:⟩} {⟨colons:⟩} {⟨stopped:⟩} {⟨alter:⟩}
  {⟨scan:w⟩} ⟨input stream⟩
```

Start a scanning branch section by executing the ⟨*preamble:*⟩ instructions. The six next arguments are function bodies used to define the corresponding branch functions. The ⟨*scan:w*⟩ instructions are finally executed. They are expected to scan the input stream and terminate by calling one of the branch functions just defined. They consist of some `..._scan_...` function defined below.

```
\cs_new:Npn \BNVS_scan:nnnnnnnnw #1 #2 #3 #4 #5 #6 #7 #8 {
  #1
  \BNVS_scanned_close_set:n     { #2 }
  \BNVS_scanned_comma_set:n     { #3 }
  \BNVS_scanned_one_colon_set:n { #4 }
  \BNVS_scanned_colons_set:n    { #5 }
  \BNVS_scanned_stop_set:n      { #6 }
  \BNVS_scanned_alter_set:n     { #7 }
  #8
```

There must be absolutely no code here. No hooking allowed either.

```
}
```

\BNVS_begin_scan:

```
\BNVS_begin_scan:
```

Start a scanning group.

```
\cs_new:Npn \BNVS_begin_scan: {
  \BNVS_scan_will_begin:
  \BNVS_begin:
  \BNVS_scan_did_begin:
}

\cs_new:Npn \BNVS_scan_will_begin: {
  \BNVS_xprt_stored:
}

\cs_new:Npn \BNVS_scan_did_begin: {
  \BNVS_xprt_clear:
  \BNVS_xprt_wrap_prefix:n {}
  \BNVS_next_set:n {}
}
```

\BNVS_end_scan:n
\BNVS_end_scan:
\BNVS_end_scan_next:

\BNVS_end_scan:n {⟨material-x⟩}
\BNVS_end_scan:
\BNVS_end_scan_no_I:
\BNVS_end_scan_next:

One of these functions must be called to close the TEX scan group. \BNVS_end_scan_next: calls \BNVS_end_scan: and what was stored as next instructions before closing the group. \BNVS_end_scan: call \BNVS_end_scan:n on what was exported before closing the group (at this level), once prepared by \BNVS_xprt_wrapped:n... Notice that ⟨material-x⟩ is x-expanded.

```
\cs_new:Npn \BNVS_end_scan:n #1 {
  \__bnvs_tl_if_empty:cTF { I } {
    \exp_args:NNx \BNVS_end: \BNVS_xprt:n {
      \BNVS_xprt_wrapped:n { #1 }
    }
  } {
    \BNVS_tl_use:nvv {
      \exp_args:Nx \BNVS_end_scan:nnn {
        \BNVS_xprt_wrapped:n { #1 }
      }
    } { I } { S ( \l__bnvs_I_tl ) }
  }
}
\cs_new:Npn \BNVS_end_scan:nnn #1 #2 #3 {
  \BNVS_end:
  \BNVS_xprt:n { #1 }
  \__bnvs_tl_set:cn I { #2 }
  \__bnvs_tl_set:cn { S ( #2 ) } { #3 }
}

\cs_new:Npn \BNVS_end_scan: {
  \BNVS_xprt_use:n {
    \BNVS_end_scan:n
  }
}

\cs_new:Npn \BNVS_end_scan_next: {
  \BNVS_next_use:n { \BNVS_end_scan: }
}
```

\BNVS_begin_scan:nn

\BNVS_begin_scan:nn {⟨xprt:n⟩} {⟨next:⟩}

OBSOLETE. Call \BNVS_begin_scan: with default arguments that end the scan group and call the eponym branch function.

```
\cs_new:Npn \BNVS_begin_scan:nn #1 #2 {
  \BNVS_begin_scan:
  \BNVS_xprt_wrap:n { #1 }
  \BNVS_next_set:n { #2 }
}
```

\BNVS_scan:nnw

\BNVS_scan:nnw {⟨preamble:⟩} {⟨scan:w⟩} ⟨input stream⟩

Start a scanning branch section by executing the ⟨preamble:⟩ instructions, setting the branch function to default values and executing the ⟨scan:w⟩ instructions.

```
\cs_new:Npn \BNVS_scan:nnw #1 #2 {
  \BNVS_scan:nnnnnnnnw {
    \BNVS_begin_scan:
    #1
  } {
    \BNVS_end_scan: \BNVS_scanned_close:w
  } {
    \BNVS_end_scan: \BNVS_scanned_comma:w
  } {
    \BNVS_end_scan: \BNVS_scanned_one_colon:w
  } {
    \BNVS_end_scan: \BNVS_scanned_colons:w
  } {
    \BNVS_end_scan: \BNVS_scanned_stop:
  } {
    \BNVS_end_scan: \BNVS_scanned_alter:w
  } {
    #2
  }
}
```

1.3.11 Delimiters

We assume that delimiters are paired and balanced. Nevertheless, some errors are catched and properly managed.

`\BNVS_scan_close:nnnnNn` `\BNVS_scan_close:nnnnNn {⟨xprt prefix:w⟩} {⟨xprt wrap:n⟩} {⟨close:w⟩} {⟨stop:⟩} ⟨close⟩ {⟨scan:w⟩}`

An opening delimiter has been scanned, scan the material up to the balancing closing delimiter using ⟨*scan:w*⟩. Implementation detail: we begin a scanning group that will be closed on scanning the ⟨*close*⟩ delimiter, or a different closing delimiter, allways at the same level. The ⟨*scan:w*⟩ function scans the ⟨*input stream*⟩, and is expected to end on a closing delimiter at the current level by sending `\BNVS_scanned_close:w`. This is the normal instruction flow.

```
\cs_new:Npn \BNVS_scan_close:nnnnNn #1 #2 #3 #4 #5 #6 {
  \BNVS_scan:nnnnnnnnw {
```

One group level for the contents.

```
    \BNVS_begin_scan:
    \cs_set:Npn \BNVS_scanned_close:Nw ##1 {
      \tl_if_eq:nnF { #5 } { ##1 } {
```

If the closing delimiter scanned is not the expected one, raise.

```
        \BNVS_error:n { Unexpected~`##1'~instead~of~`#5'. }
      }
```

Anyways, branch here.

```
      \BNVS_scanned_close:w
    }
    \BNVS_xprt_wrap_prefix:n { #1 }
    \BNVS_xprt_wrap_raw:n { #2 }
  } {
```

- Close branch. End the group and execute ⟨*close:w*⟩ instructions. Intermediate groups should be closed appropriately with the right definition of branch functions. Clearing the `I` variable before returning has the side effect not to export the actual value of `I` and keep what is already in the containing group.

```
    \__bnvs_tl_clear:c I
    \BNVS_end_scan:
    #3
  } {
```

- Any other branch function raises and forwards to `\BNVS_scanned_close:w`,

```
    \BNVS_error:n { Missing~`#5'. }
    \BNVS_scanned_close:w
  } {
    \BNVS_error:n { Missing~`#5'. }
    \BNVS_scanned_close:w
  } {
    \BNVS_error:n { Missing~`#5'. }
    \BNVS_scanned_close:w
  } {
```

- or `\BNVS_scanned_stop:` after the group ends.

```
    \BNVS_error:n { Missing~`#5'. }
    \BNVS_end_scan:
    #4
  } {
```

- Unreachable code?

```
    \BNVS_error:n { Missing~`#5'. }
    \BNVS_scanned_close:w
  } {
```

- One group level for the contents.

```
    #6
  }
}
```

`\c__bnvs_open_rnd_regex` Round closing delimiter.

```
\regex_const:Nn \c__bnvs_open_rnd_regex { \s* \( %)
  \s* }
```

(*End of definition for* `\c__bnvs_open_rnd_regex`.)

`\c__bnvs_open_sqr_regex` Square closing delimiter.

```
\regex_const:Nn \c__bnvs_open_sqr_regex { \s* \[ %]
  \s* }
```

(*End of definition for* `\c__bnvs_open_sqr_regex`.)

\c__bnvs_open_crl_regex Round closing delimiter.

```
\regex_const:Nn \c__bnvs_open_crl_regex { \s* \{ %}
  \s* }
```

(*End of definition for* `\c__bnvs_open_crl_regex`.)

\BNVS_scan_inside:w `\BNVS_scan_inside:w` ⟨*input stream*⟩

Scan balanced material taking only delimiters into account. Used to parse calls to functions.

```
\cs_new:Npn \BNVS_scan_inside:n #1 {
  \BNVS_xprt:n { #1 }
  \BNVS_scan_inside:w
}

\group_begin:
\char_set_catcode_group_begin:N <
\char_set_catcode_group_end:N >
\char_set_catcode_other:N \{
\char_set_catcode_other:N \}
\cs_new:Npn \BNVS_scan_inside:w <
  \peek_regex_replace_once:NnTF \c__bnvs_open_rnd_regex < > <
    \BNVS_scan_close:nnnnNn
      < > < ##1 >%(
      < \BNVS_xprt:n < ) > \BNVS_scan_inside:w >%(
      < \BNVS_error:n < Missing~`)' > %(
        \BNVS_xprt:n < ) > \BNVS_scan_stop: > %(
      ) < \BNVS_xprt:n < ( > \BNVS_scan_inside:w >%)
  > <
    \peek_regex_replace_once:NnTF \c__bnvs_open_sqr_regex < > <
      \BNVS_scan_close:nnnnNn
        < > < ##1 >%[
        < \BNVS_xprt:n < ] > \BNVS_scan_inside:w >%[
        < \BNVS_error:n < Missing~`]' > %[
          \BNVS_xprt:n < ] > \BNVS_scan_inside:w >%[
        ] < \BNVS_xprt:n < [ > \BNVS_scan_inside:w >%]
    > <
      \peek_regex_replace_once:NnTF \c__bnvs_open_crl_regex < > <
        \BNVS_scan_close:nnnnNn
          < > < ##1 >%{
          < \BNVS_xprt:n < } > \BNVS_scan_inside:w >%{
          < \BNVS_error:n < Missing~`}' > %{
            \BNVS_xprt:n < } > \BNVS_scan_inside:w >%{
          } < \BNVS_xprt:n < { > \BNVS_scan_inside:w >%}
      > <
        \BNVS_scan_close:Fw <
          \BNVS_scan_stop:F <
            \BNVS_scan_inside:n
          >
        >
      >
    >
  >
>
\group_end:
```

1.3.12 Comma separated lists

Here is a partial grammar for a comma separated list template.

```
⟨csl(⟨item⟩)⟩ ::= ⟨item⟩? ( ⟨comma⟩ ⟨item⟩ )*
```

The corresponding AST template is below, where ⟨*item*⟩*is the AST for ⟨*item*⟩:

```
\:nn { ⟨item⟩* } { \:nn { ⟨item⟩* } { ⟨...⟩\:nn { ⟨item⟩* } { } { } ⟨...⟩ } }
```

`\BNVS_scan_csl:nnnnw` `\BNVS_scan_csl:nnnnw {⟨preamble:⟩} {⟨close:w⟩} {⟨stop:⟩} {⟨scan:w⟩} ⟨input stream⟩`

This function scans the comma separated list of items, regardless the enclosing delimiters. It begins a scanning TEX group that is ended by a call to `\BNVS_scanned_close:w`, executing ⟨*close:w*⟩, or `\BNVS_scanned_stop:`, executing ⟨*stop:*⟩. The scanned material is then exported between braces. On scanning a comma, a new group is created. The exported material is empty for an empty list, for input `a` it is `\:nn{a}{}`, for input `a,b` it is `\:nn{a}{\:nn{b}{}}`, and so on.

```
\cs_new:Npn \BNVS_scan_csl:nnnnw #1 #2 #3 #4 {
  \BNVS_scan:nnnnnnnnw {
```

One group level for the contents.

```
    \BNVS_begin_scan:
    \BNVS_xprt_wrap_prefix:n { \:nn }
    \BNVS_xprt_wrap:n { { ##1 } { } }
```

Execute the ⟨*preamble:*⟩ instructions..

```
    #1
  }
```

Close branch.

```
    { \BNVS_end_scan: #2 }
    {
      \BNVS_xprt_use:n { \tl_if_empty:nTF } {
```

Comma branch, the first item is empty, skip it.

```
        #4
      } {
```

Comma branch, the first item is not empty, there will be another item, possibly empty.

```
        \BNVS_xprt_wrap:n { { ####1 } }
        \BNVS_end_scan:
        \BNVS_scan:nnnnnnnnw {
          \BNVS_begin_scan:
```

```
            \BNVS_xprt_wrap:n { { ####1 } }
          }
            { \BNVS_end_scan: \BNVS_scanned_close:w }
            { \BNVS_error:n { Unreachable } \BNVS_scanned_close:w }
            { \BNVS_error:n { Unreachable } \BNVS_scanned_close:w }
            { \BNVS_error:n { Unreachable } \BNVS_scanned_close:w }
            { \BNVS_end_scan: \BNVS_scanned_stop: }
            { #4 }
          {
            \BNVS_scan_csl:nnnnw { } {
```

Close branch.

```
              \BNVS_if_xprted_empty:TF {
                \BNVS_end_scan:
                \BNVS_xprt:n { { } }
              } {
                \BNVS_end_scan:
              }
              \BNVS_scanned_close:w
            } {
```

Stop branch.

```
              \BNVS_if_xprted_empty:TF {
                \BNVS_end_scan:
                \BNVS_xprt:n { { } }
              } {
                \BNVS_end_scan:
              }
              \BNVS_scanned_stop:
            } {
              #4
            }
          }
        }
      }
      { \BNVS_error:n { Unreachable } \BNVS_scanned_close:w }
      { \BNVS_error:n { Unreachable } \BNVS_scanned_close:w }
      { \BNVS_end_scan: #3 }
      { \BNVS_error:n { Unreachable } \BNVS_scanned_close:w }
```

By default, the list contains only one item.

```
    { #4 }
}
```